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Measuring capacitance & ESR

Last Modification: January 20, 2014

There are various ways to determine the capacitance of capacitors. This article describes a number of measurement methods. Also the ESR (equivalent series resistance) can be measured with most of the presented methods.

A capacitor has beside the most important property; the capacitance, also parasitic properties. The most important of these are the series resistance and the self-inductance who is also connected in series with the capacitance. The model of a capacitor with its parasitic components is shown in the figure on the right. This model doesn't include the in parallel connected leakage resistance. In most cases these are negligible, but if this parameter is necessary, it can be measured with a DC-resistant measurement.

The following measurement methods measure the capacitance and ESR. To determine also the parasitic self-inductance another measurement method must be used described in the article [Parasitic properties](#).

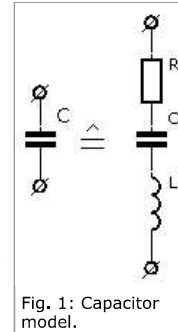


Fig. 1: Capacitor model.

Measuring with a squarewave

By connecting a capacitor to a squarewave generator a typical voltage shape across the capacitor arises. By analyzing the voltage shape the capacity and internal series resistance can be determined.

Measurement arrangement

Figure 2 shows the measurement arrangement. The capacitor to test is directly connected to the output terminals of the function generator who's delivering a squarewave voltage. The voltage across the capacitor is measured with an oscilloscope. The function generator is set at the maximum output voltage, and the frequency is adjusted so that the voltage across the capacitor is kept at a low level. In this way almost the entirely voltage is dropped across the internal generator resistance. It is like the capacitor is connected to a current source. The current that flows is calculated as:

$$I := \frac{V_g}{R_i} = \frac{10 \text{ Vt}}{50 \Omega} = 200 \text{ mA}$$

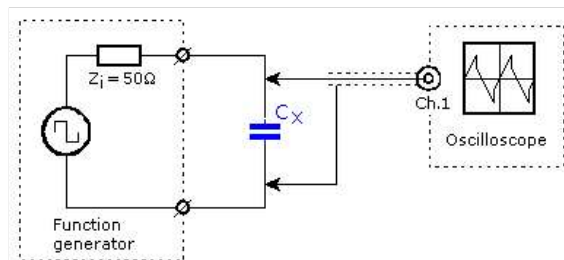


Fig. 2: Squarewave voltage capacitance measurement circuit.

The voltage V_g is the open terminal top value of the generator. In most cases the generator voltage is given as a top-top value at a specific load, often 50 Ω . The unloaded top value is therefore equal to the loaded top-value.

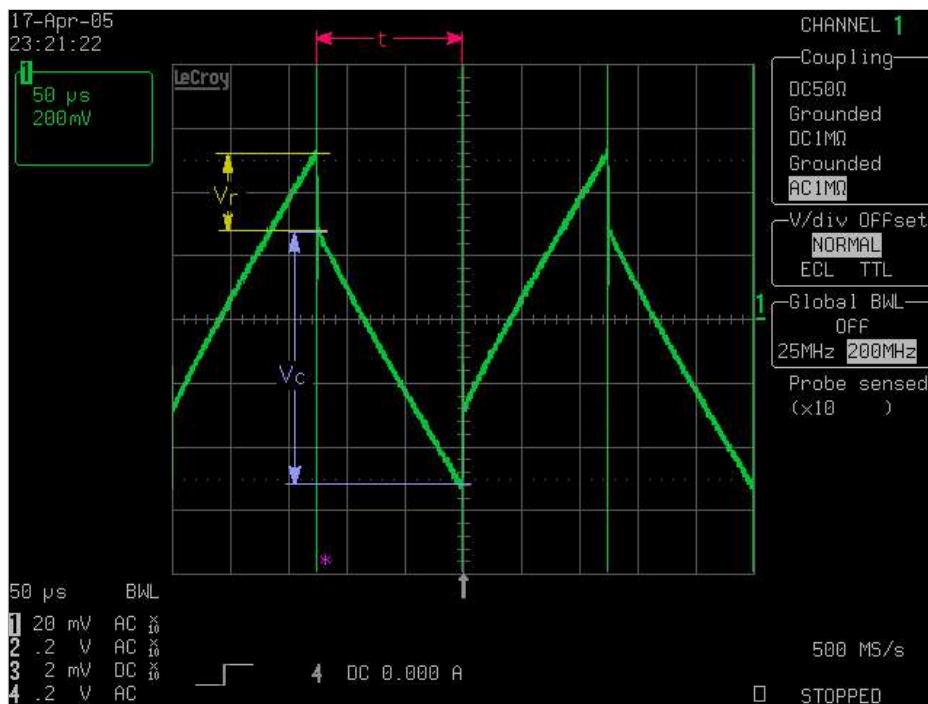


Fig. 3: Oscilloscope screen shot of a capacitance measurement on a 33 μF , 63 V electrolytic capacitor with a squarewave voltage.

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Multimeters

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Capacitance

The screen shot in figure 3 shows linear increasing and decreasing voltages. The amplitude of the slopes is marked in a blue color and labeled V_C . This voltage change occurs during the time t , marked in red. With these numbers the capacitance can be calculated:

$$C := \frac{I \cdot t}{V_C} = \frac{0.2 \text{ A} \cdot 125 \cdot 10^{-6} \text{ s}}{0.82 \text{ V}} = 30.4 \mu\text{F}$$

Internal resistance (ESR)

The yellow arrow represents the step voltage V_R which is caused by the capacitor internal series resistance under the influence of current polarity reversal. The current changes here from -0.2 A to $+0.2 \text{ A}$, thus a current step of 0.4 A . With this information the ESR, or equivalent series resistance, can be calculated:

$$R_C := \frac{V_R}{2 \cdot I} = \frac{0.24 \text{ V}}{2 \cdot 0.2 \text{ A}} = 0.6 \Omega$$

Self-inductance

The self-inductance can't be determined with this method. However, the effects of the inductance are noticeable as spikes in the oscilloscope image. One of these spikes is marked with an asterisk.

Measuring with a sinewave

With this measurement method the capacitor is inserted in a half bridge configuration which is connected to a sinewave generator. By the measured voltages and phase difference the capacity and ESR can be determined.

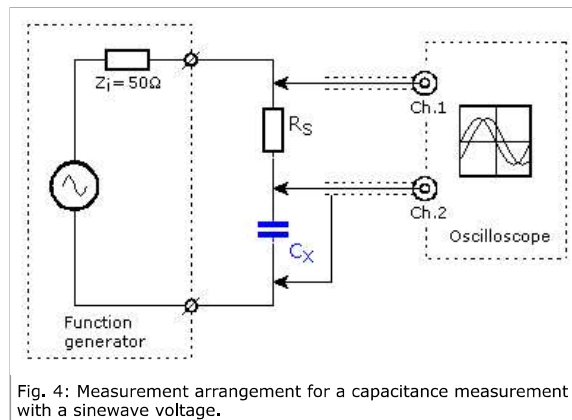
Limitations

Capacitors can almost be considered as ideal components. The equivalent series resistance is normally very small as well as the capacity in most cases. To measure the capacity and the internal resistance accurate, the measure frequency must be chosen so that the reactance and resistance are approximately the same. The phase difference between the capacitor voltage and the voltage across the reactance is then circa 45° . This means that the measure frequency must be very high in some cases, tens to hundreds megahertz. The self-inductance at these frequencies is very dominant what makes the measurement worthless. Only electrolytic capacitors who have a relative high capacity in combination with a high ESR, the measure frequency can be kept low enough to be usable. When measuring other capacitors the frequency must be chosen lower than desired what means that only the capacitance can be measured.

Two examples are given: The first one is for measuring only the capacitance, and the second one is for measuring the capacity as well as the ESR.

The measuring arrangement

The measurement arrangement is shown in figure 4. This is a half bridge configuration consisting of the resistor R_s and the unknown capacitor C_x . The bridge voltage and the voltage at the junction is measured with a two channel oscilloscope. Also the phase difference between these voltages is measured. To measure the internal resistance accurately the channel 2 probe must be placed as close as possible at the capacitor. The resistor R_s must have approximately the same value as the impedance of the capacitor.



Method 1: Measuring capacitance

The first method describes the measurement of small capacitors whereof the series resistance is negligible.

Mathematical model

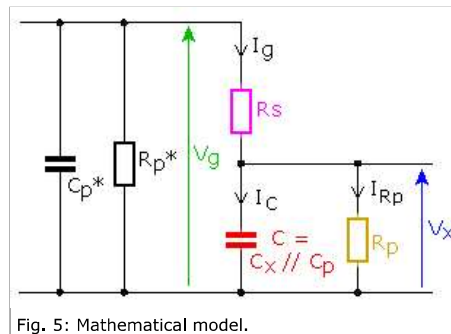
Figure 5 shows the capacitor model that is used for the calculation of the capacity C_x . The associated vector diagram is shown in figure 6. The reactance of the capacitor is relative high in this measurement. The ohmic series resistance breaks down into nothingness and is therefore not included into the model.

Parallel to the capacitor under test is the probe connected represented by the capacity C_p and the ohmic resistance R_p . The probe capacity C_p and the unknown capacitor C_x are taken together as one replacement capacity C . Because C_p (and also R_p) are known it is easy to figure out the unknown capacity.

The current is measured with the aid of R_s . The value of this resistance should be in the proximity of the reactance of the capacitor to be measured and is thus dependent on the measurement frequency, and the capacitance:

$$R_s \approx X_C := \frac{1}{2 \cdot \pi \cdot f \cdot C}$$

For a capacity of 100 pF and a measure frequency of 50 kHz , R_s must be approximately $33 \text{ k}\Omega$. For an unknown capacity the resistance R_s has to be determined empirically. The reactance is set by manipulating



the frequency. The probe for measuring the terminal voltage of the generator V_g does load the function generator but has no further influence on the measurement. This probe is represented by C_p^* and R_p^* .

Processing measurement data

Figure 7 shows the result of a measurement to an 100 pF capacitor. On the basis of this measurement is shown how the results should be processed. The following values are measured: the generator voltage V_g (5.076 V), the voltage across the capacitor V_x (3.242 V), the phase angle between these two voltages α (48.89 °) and the frequency f (50 kHz). The resistor value R_s is 33 kΩ, and the probe capacitance C_p is 12 pF.

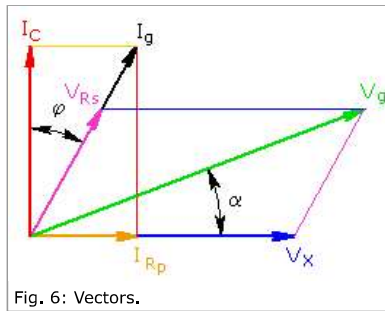


Fig. 6: Vectors.

With these numbers the voltage across the shunt resistor R_s is calculated:

$$V_{R_s} := \sqrt{(V_g \cdot \cos(\alpha) - V_x)^2 + (V_g \cdot \sin(\alpha))^2} = 3.826 \text{ V}$$

And the generator current is therefore:

$$I_g := \frac{V_{R_s}}{R_s} = 115.93 \text{ } \mu\text{A}$$

The loss angle ϕ (with respect to the parallel resistor):

$$\phi := \arccos\left(\frac{V_g \cdot \sin(\alpha)}{V_{R_s}}\right) = 1.43 \text{ deg}$$

The reactance of the total capacitance:

$$X_C := \frac{V_x}{I_g \cdot \cos(\alpha)} = 27.974 \text{ k}\Omega$$

The total capacitance is:

$$C_t := \frac{1}{2 \cdot \pi \cdot f \cdot X_C} = 113.789 \text{ pF}$$

The probe capacitance is subtracted from this total to calculate the unknown capacitance:

$$C_x := C_t - C_p = 101.789 \text{ pF}$$

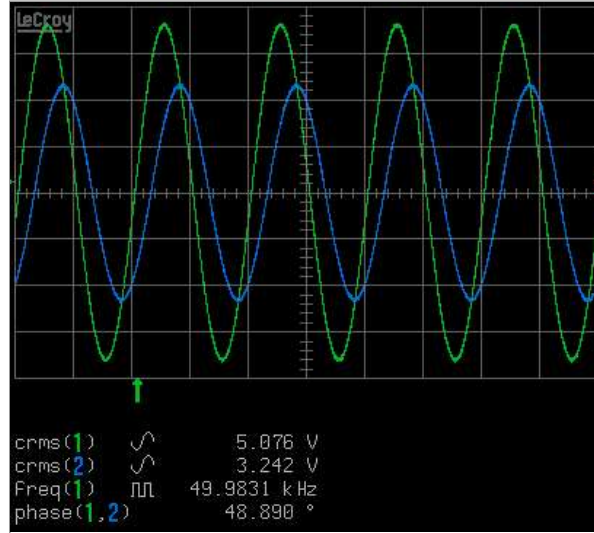


Fig. 7: Measuring a 100 pF ceramic capacitor.

Method 2: Measuring capacitance and internal resistance (ESR)

The second method describes a measurement that is suitable for measuring on larger capacities and can also determine the internal series resistance (ESR). This method is thus mainly suitable for measuring on electrolytic capacitors.

Mathematical model

Figure 8 shows the mathematical model with the associated vector diagram in figure 9. C_x and R_x are the capacitive and resistive parts of the measured capacitor. With the use of the resistor R_s the current is measured. This resistor must have a low ohmic value to keep the circuit at a low impedance. This is necessary in order to be able to determine the internal resistance. The probe capacitance and resistance are ignored.

Processing measurement data

On the basis of a measurement on a 100 μF, 16 V electrolytic capacitor is demonstrated how the capacitance and ESR can be calculated with the measured parameter. The scope picture in figure 10 shows the measurement. The measured generator voltage V_g is 417 mV, the capacitor voltage V_x is 291 mV, the phase difference between these two voltages α is 31.18 ° and the frequency f is 700 Hz. A 2.2 Ω resistor is used for R_s .

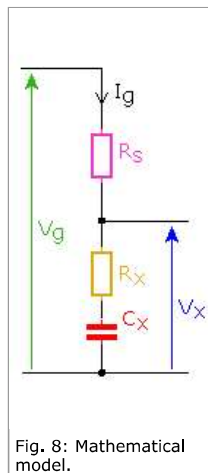


Fig. 8: Mathematical model.

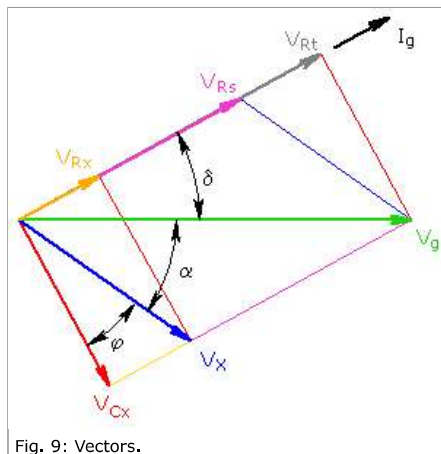


Fig. 9: Vectors.

With the measured voltages V_g and V_x and the phase shift α the voltage across the resistor R_s is calculated:

$$V_{R_s} := \sqrt{(V_g - V_x \cdot \cos(\alpha))^2 + (V_x \cdot \sin(\alpha))^2} = 0.235 \text{ V}$$

And therefore the current through the circuit:

$$I_g := \frac{V_{R_s}}{R_s} = 0.107 \text{ A}$$

The loss angle is:

$$\phi := 90 \text{ deg} - \left(\alpha + \arccos \left(\frac{V_g - V_x \cdot \cos(\alpha)}{V_{R_s}} \right) \right) = 14.262 \text{ deg}$$

The voltage across the capacitive part:

$$V_{C_x} := V_x \cdot \cos(\phi) = 0.282 \text{ V}$$

The reactance:

$$X_{C_x} := \frac{V_{C_x}}{I_g} = 2.635 \Omega$$

The capacity:

$$C_x := \frac{1}{2 \cdot \pi \cdot f \cdot X_{C_x}} = 86.285 \mu\text{F}$$

The voltage across the ohmic part:

$$V_{R_x} := V_x \cdot \sin(\phi) = 0.072 \text{ V}$$

And the ESR of the capacitor is :

$$R_x := \frac{V_{R_x}}{I_g} = 0.67 \Omega$$

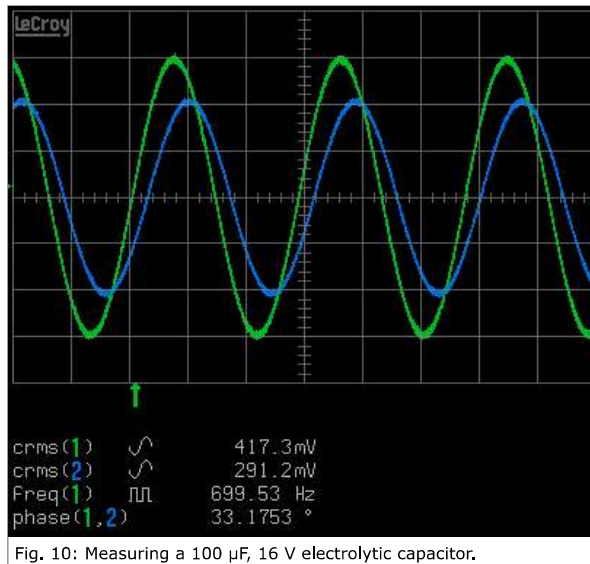


Fig. 10: Measuring a 100 μF , 16 V electrolytic capacitor.

There is a side note with this example: The loss angle ϕ which has a value of 14° that makes the difference between the reactance and ohmic resistance relative high. This has the effect that the capacitance is calculated with a greater accuracy than the internal series resistance. If one wants to determine the ESR with a greater accuracy than the measure frequency has to be adjusted to make the loss angle larger.

Resonance measurement

Another way to measure the capacitance is to include the unknown capacitor in a resonance circuit. The accuracy is directly dependent on the used reference inductor. Inductors with a small tolerance are rare and expensive.

Measurement arrangement

The unknown capacitor is connected in series with a reference inductor and connected to a sinewave generator. The voltage across this resonance circuit is measured with an AC-voltmeter. The most voltmeters are not suited to measure very low or high frequencies. In most cases a peak detector can fulfill this task.

If the resonance circuit has a high DC-resistance it is possible that the resonance frequency can't be found. In this case a extra resistor R_a can be placed in series with the generator.

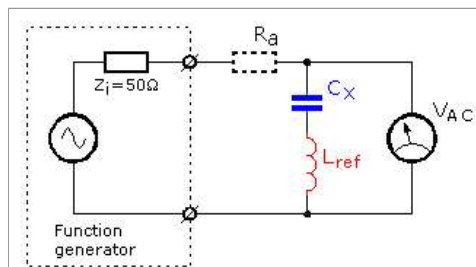


Fig. 11: Resonance method measuring arrangement for capacitors.

The measurement

With the sinewave generator the frequency range where the resonance frequency is expected is slowly swept. The resonance frequency is noticeable by a sharp voltage drop. If the sinewave generator doesn't produce a clean signal the chance exists that a harmonic causes a weaker dip. It is advisable to continue looking for a more distinctive dip.

With the found resonance frequency f_0 the unknown capacitance can be calculated:

$$C_x := \frac{1}{(2 \cdot \pi \cdot f_0)^2 \cdot L_{\text{ref}}} \quad [\text{F}]$$

The internal series resistance can't be determined with a great accuracy. Even if the ohmic resistance of the inductor is well known, the various junction resistances are still unknown.

Accuracy

Influence cables and instruments

As noted previously, the measurement is affected because it is connected to measuring equipment. For a more accurate measurement the model of the measurement arrangement has to be added to the capacitor model, where the self-inductions, resistances and capacitances of the measurement instruments and cables is taken into account. This is particularly important at high measuring frequencies.

Dielectric losses

The dielectric of capacitors is not ideal. It involves losses appear as ohmic losses. The dielectric losses are also frequency and voltage dependent.

Measuring electrolytic capacitors

Be aware of the polarity when measuring electrolytic capacitors there should be no AC voltage present. Voltages that become negative with respect to the capacitors polarity may cause faulty results and may damage the capacitor. An offset voltage can in most cases be included to the sinewave of a function generator. In this way can be ensured that never the wrong polarity is applied to the capacitor.

Parasitic self-induction

The described measuring methods cant measure the parasitic self-inductance. The article [Parasitic properties](#) how this and the frequency dependent ohmic resistance is measured.

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At 18 jan 2017, 21:13:56 wrote [Freddy](#).

If you measure the voltage across Rs with a voltmeter you'll get Vrs. But by only using a voltmeter you'll lose the phase relation between the voltage across and the current trough the capacitor. You need the phase to determine both the capacitance and ESR.

At 13 feb 2017, 18:50:39 wrote [Glenn](#)

Very useful article.
 A couple of corrections on Method1:
 The VRs equation should have the second square term outside the brackets, i.e.: $(V_g \sin(a))^2$
 The Reactance calculation should be "sin(phi)", not "sin(alpha)".

At 13 feb 2017, 18:52:44 wrote [Glenn](#)

Very useful article.
 A couple of corrections on Method1:
 The VRs equation should have the second square term outside the brackets, i.e.: $(V_g \sin(a))^2$
 The Reactance calculation should be "sin(phi)", not "sin(alpha)".

At 08 jul 2017, 11:34:47 wrote [Jeremy](#).

I'm interested to know how you came to 14.262 degrees when determining phi. My calculation of $(V_g - V_x \cos(\alpha))/V_g$ yields 0.403. Plugging that into acos gives 66.24 degrees. What number did you use for acos?

At 15 sep 2018, 00:40:26 wrote [BrianE](#)

Method one reduces to the following (solving the nodal equation directly):
 $V_g \sin(\alpha) / (V_x^2 \pi f R_s) - \text{probe capacitance}$

At 21 apr 2019, 01:46:35 wrote [Daniel](#)

I'd like to measure the distance of two narrow metallic plates with size around 1cm^2 and 1cm^2 at a frequency of at least 100Hz, preferably rather around 1-10 kHz. I think measuring the capacity is the best way to measure the distance. (I have a good microscope to measure the precise distance, but is very slow. However, it is suitable to calibrate a fast method.) Which of these method do you suggest to measure the capacity? Or none of them?

At 22 apr 2019, 12:12:59 wrote [Freddy](#)

Yes, it is certainly possible to determen the distance of the plates by measuring the capacitance. But be aware that the capacitance is very small: at a distance of 0.1 mm the capacitance is only 8.85 pF! 100 Hz or even 10 kHz is too low to do it reliable. Even at 1 MHz the reactance is 18 kΩ, but feasible.

At 21 mei 2019, 17:05:55 wrote [Mykeil](#)

Hi. Great tutorial. I found small mistake.
 Sin(phi) should be in a formula of Xc calculation in Method 1: Measuring capacitance.
 Otherwise it will not give 27.974kOhm, as written.
 Thank you, Regards :)

At 15 apr 2022, 01:46:41 wrote [Lawrence](#)

May I ask if ESR is included invisibly in the capacitance measurement

At 15 apr 2022, 11:45:03 wrote [Freddy](#)

Methode 1 & 2 obtain the capacitance and ESR independent of each other.
 Other type of measurements or with dedicated capacitance measurement devices you can't be sure if (unless you're sure how it works) the ESR influences the measured capacitance value.

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